

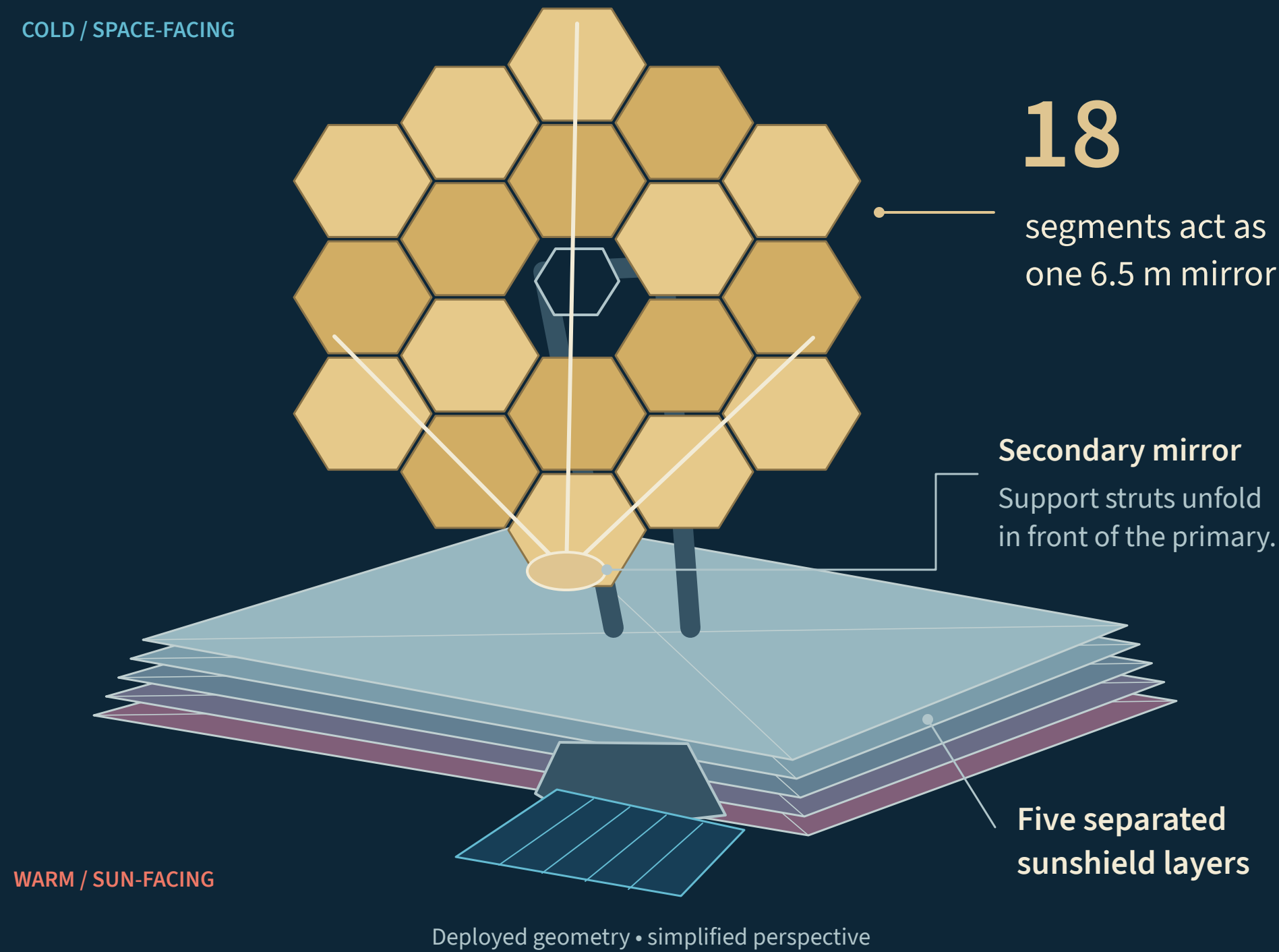
Webb: unfold, cool, focus

The engineering behind a cold infrared observatory: a deployable mirror, a thermal barrier and months of optical alignment.

01 A large mirror on the cold side

Unfolding creates an observatory that could never fit inside its rocket.

COLD / SPACE-FACING



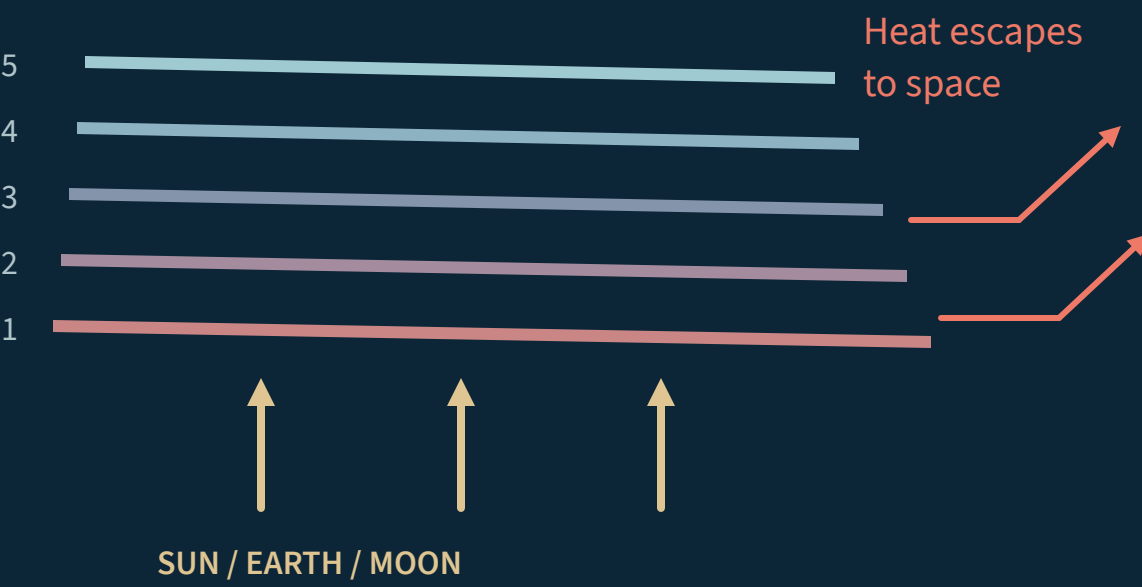
02 Keep its own heat out

An infrared telescope must stay cold enough to detect faint signals.

TELESCOPE SIDE

Below 50 K

Passive cooling after deployment



Vacuum gaps limit conduction; separated membranes radiate heat outward.

MIRI needs an extra cooler

Its mid-infrared detectors operate near 7 K, colder than passive cooling alone can provide.

03 Deploy first. Then turn eighteen images into one.

Physical deployment ends before optical commissioning is complete.

